



INTERNATIONAL ISLAMIC UNIVERSITY
ISLAMABAD
Department of Mathematics & Statistics

To Whom It May Concern

It is my great pleasure to write this letter of recommendation in support of Mr. Muhammad Nazim for admission to a PhD program in your esteemed institution. I have known Mr. Nazim during his graduate studies at the Department of Mathematics and Statistics, Faculty of Sciences at International Islamic University Islamabad, Pakistan.

Mr. Nazim is an exceptionally motivated and talented young researcher with a strong academic background in applied mathematics, numerical analysis, and machine learning techniques for solving complex mathematical and engineering problems. During his master's studies, he demonstrated outstanding analytical ability, intellectual curiosity and dedication to research.

His master's thesis titled "*Analysis of Parameter Uncertainty on Stochastic Heat Transfer Problem*" reflects his deep understanding of stochastic modeling, uncertainty quantification, and numerical simulation techniques.

Mr. Nazim has actively contributed to research in areas such as stochastic modeling, heat and fluid flow analysis and machine learning-based solutions for nonlinear partial differential equations. He has authored and co-authored several research papers submitted to reputable international journals, including work on hybrid stochastic-deterministic frameworks, uncertainty propagation, and Physics-Informed Neural Networks for complex fluid flow systems. His research demonstrates strong technical expertise in TensorFlow, MATLAB, Mathematica, and Python, as well as solid knowledge of Monte Carlo methods, Polynomial Chaos Expansion, and advanced numerical techniques.

Beyond his technical abilities, Mr. Nazim is a disciplined, hardworking, and highly responsible individual. He consistently shows perseverance in addressing complex research problems and possesses the patience required for rigorous scientific investigation. He is capable of working independently while also contributing effectively within collaborative research environments.

What distinguishes Mr. Nazim is his strong passion for advancing mathematical research and applying mathematical tools to solve real-world scientific and engineering problems. His interests in stochastic simulations, machine learning methods for nonlinear partial differential equations, and numerical analysis place him among promising young researchers in applied mathematics.

Based on my experience working with him, I strongly believe that Mr. Muhammad Nazim has the intellectual capability, research potential, and determination required to successfully complete a PhD program and contribute meaningfully to the global research community. I strongly recommend him for admission to your PhD program without reservation.

Please feel free to contact me if further information is required.

Sincerely,

Prof. Dr. Ahmed Zeeshan

Department of Mathematics and Statistics

International Islamic University Islamabad, Pakistan

Email: ahmad.zeeshan@iiu.edu.pk

Phone: +92-51-9019759